On the Economics of Cultural Transmission: Welfare Analysis*

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Abstract

We study the welfare properties of cultural transmission processes governed by parental socialization choices and social interactions as well as mediated by cultural leaders. We show that, at equilibrium, parents generally under-supply socialization effort, while strategic complementarities induce cultural leaders to over-supply it.

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1 Introduction

We study the welfare properties of cultural evolution by comparing decentralized socialization choices of culturally heterogeneous groups — made by parents or cultural leaders — with socially efficient dynamics.

Cultural heterogeneity - especially when manifested along their ethnic or religious dimension - has arguably important effects on socio-economic outcomes. For instance, a large literature has documented a negative association between ethnic fractionalization and trust, social capital, cooperation and coordination in collective action, inducing in turn negative effects on public goods provision, growth, and government quality.[1] Relatedly, immigration flows are generally associated to perceived cultural externalities imposed on natives, exacerbated by a perceived sluggishness of the cultural integration process.[2]

The socio-economic effects of cultural heterogeneity motivates our present theoretical study of the efficiency properties of cultural evolution - that is, of the process underlying the manifestation of cultural heterogeneity. We conceptualize cultural evolution as an equilibrium outcome of choices governed by i) inter-generational parental socialization ('direct vertical socialization'), by ii) social interactions ('oblique and horizontal socialization'), as well as by iii) cultural leaders or institutions concerned e.g., with ethnic or religious policies.[3] In this context, we characterize the welfare properties of cultural evolution: we identify and sign the systematic distortions on the supply of socialization effort across agents which lead to inefficient dynamics of the distribution of cultural traits in a population.

A related literature studies cultural integration - in the context of immigration, stressing its

[1] See Easterly and Levine (1997); Alesina, Baqir and Easterly (1999); La Porta et al. (1999); Alesina and La Ferrara (2005); .

[2] See e.g., Algan et al. (2012); Abramitzky and Boustan (2017); Bisin and Tura (2019). Indeed, existing evidence documents that immigrants actively preserve and transmit their cultural identity across generations: see e.g., Abramitzky, Boustan and Eriksson (2020); Fouka (2020) on use of home languages and ethnic naming conventions, Gordon (1964); Meng and Gregory (2005); Furtado and Trejo (2013); Bisin and Tura (2019); Guirkinger, Platteau and Wahhaj (2019) on intermarriage patterns, Manning and Roy (2010) on self-reported national identity, and Hwang (2019) on residential clustering. The salience of the cultural identity across immigrant groups living in the same host country on a wide variety of outcomes is documented by Fernández and Fogli (2006, 2009); Fernández (2011); Giuliano (2007); Alesina, Giuliano and Nunn (2013). Economic effects in the labor market are instead arguably small: see Card (1990), Dustmann, Schönberg and Stuhler (2017), and Caiumi and Peri (2024) and Borjas (2003) for a dissenting opinion.

[3] This approach to cultural transmission has been introduced by Bisin and Verdier (2001), building on the contributions of Cavalli-Sforza and Feldman (1981) and Boyd and Richerson (1988) in evolutionary biology and anthropology; see Bisin and Verdier (2025*a*) for a recent survey. Children's identity formation choices - introduced in economics by Akerlof and Kranton (2000) and recently surveyed in Akerlof and Kranton (2010, 2026) - are studied in this context as well; see e.g., , Bisin et al. (2006, 2011); Bisin and Verdier (2025*b*). The canonical model of socialization by cultural leaders has been introduced by Verdier and Zenou (2018) and Almagro and Andrés-Cerezo (2020).

positive externality on natives.[4] Our analysis of welfare properties builds instead on a notion of Pareto efficiency, encoding the ex-ante preferences of parents of all cultural groups in the population. In the immigration context, externalities go symmetrically both ways, from immigrant to natives and from natives to immigrant.[5]

More specifically, in our model parents choose socialization effort taking the current and future distribution of cultural traits as given, while cultural leaders anticipate how their effort affects the future cultural distribution and internalize this effect in their decisions.[6] We show that parental socialization is generically inefficient due to its positive externality through population dynamics and social interactions that individual parents do not internalize. Interestingly, welfare improvements require policies which re-distribute socialization costs across groups, besides controlling socialization rates. We also show that cultural leaders' socialization choices are generically inefficient, due to the strategic complementarities in their choice problems. Further, we show that at equilibrium, parents generally under-supply socialization effort, while cultural leaders over-supply it. With symmetric groups, the actual path of cultural evolution is unaffected by the inefficiencies: e.g., the excess socialization that strategic complementarities generate is pure waste, sustaining the same cultural trajectory at needlessly higher cost rather than pushing assimilation or differentiation forward at an inefficient pace. Distortion of the evolutionary path itself instead arises from asymmetries: whenever one group's institutions systematically over- or under-socialize relative to the other's, its cultural dynamics are too quick or too slow relative to what a planner weighing both groups' ex-ante preferences would choose.

We conclude noticing several historical events which align anecdotally well with our result that cultural leaders over-supply socialization effort. In Ottoman Macedonia after 1878, the Greek, Bulgarian, and Serbian churches competed for the allegiance of the same peasant population by building schools, hiring teachers, and funding scholarships. Each side expanded in response to the others, and contemporaries saw the resulting school race as politically rather than educationally motivated (Brooks, 2015). In Bohemia, rival German and Czech school associations, both founded in 1880, fought a similar battle over minority schools along the language frontier, even though many of the families at stake were themselves largely indifferent to nationality (Zvánovec, 2020; Zahra,

[4]See Lazear (1999); Gabszewicz, Ginsburgh and Weber (2011); Brock et al. (2024); Kuran and Sandholm (2008); Konya (2007).

[5]See Benhabib and Jovanovic (2012); Ding and Goyal (2026); Della Lena, Sato and Zenou (2026) for welfare analyses - in different contexts - along lines more similar to ours.

[6]In the canonical formulation of the cultural transmission model in Bisin and Verdier (2001) - with no forward-looking parents and no cultural leaders - the efficiency question turns out to be un-interestingly trivial; see Footnote 16.

2008). And in rural America, Protestant denominations built competing churches in the same small communities, a duplication contemporaries criticized as "overchurching"; reformers argued that all denominations would gain by scaling back together (Gill and Pinchot, 1913, 1919; Vogt, 1921). This is exactly the joint reduction in socialization effort that, in our model, makes both groups better off relative to the cultural leaders' equilibrium (Proposition 4.3).

The rest of the paper is organized as follows. In Section 2 we introduce the canonical cultural transmission model and show that more forward-looking parents socialize more. In Section 3 we define our notions of equilibrium for parents and cultural leaders, and show that under some conditions the leaders socialize more than the parents. In Section 4, we define efficiency and show that (and characterize how) parental and cultural leader socialization is generically inefficient. Section 5 concludes.

2 Cultural transmission: The canonical model

We study the canonical economic model of cultural transmission. In this model, a population is distributed across two cultural traits, $\{a, b\}$. Let q_t^i denote the fraction of the population with cultural trait i at time t, where $t = 0, 1, \ldots, \infty$. We use i, j as indexes in $\{a, b\}$, with the understanding that $i \neq j$; hence, e.g., $q_t^j = 1 - q_t^i$.

Each parent has one child and reproduction is asexual. The parental socialization process links generations onto an overlapping societal structure.

2.1 Technology

Parents socialize their children directly to transmit their own (the parents') cultural trait. Parental effort in direct socialization is denoted τ^i. Letting P_t^{ii} (resp. P_t^{ij}) denote the probability that a parent of cultural trait i at time t has a child of cultural trait i (resp. j) at $t+1$, the socialization technology is the following:[7]

$$\begin{aligned} P_t^{ii} &= \tau_t^i + (1 - \tau_t^i) q_t^i \\ P_t^{ij} &= (1 - \tau_t^i)(1 - q_t^i) \end{aligned} \tag{1}$$

[7] The technology is straight from Cavalli-Sforza and Feldman (1981) and Boyd and Richerson (1988).

The probability that a parent of cultural trait i at time t has a grand-child of cultural trait i (resp. j) at $t+h$,

$$P^{ii}_{t,t+h}, P^{ij}_{t,t+h}, \quad h \geq 1,$$

is constructed recursively.

This cultural transmission mechanism induces an evolution of cultural traits characterized by the difference equation:

$$q^i_{t+1} - q^i_t = q^i_t(1-q^i_t)\left(\tau^i_t - \tau^j_t\right) \tag{2}$$

2.2 Socialization choice

Socialization rates τ^i_t, τ^j_t are endogenously chosen either by parents or by cultural leaders. Cultural leaders are benevolent and maximize parental utility, by cultural group. We assume - for simplicity and ease of comparison - that the choice variable is the same for parents and cultural leaders, τ^i_t, for all t; that is, we short-circuit the indirect effect of cultural leaders on parental choices via cultural institutions, like e.g., schools and churches. The fundamental distinction is that parents take as given the dynamics of q^i_t, for all t, while cultural leaders anticipate the dependence of the dynamics on their own choices. As a consequence, parents solve a simple choice problem, while cultural leaders play a non-cooperative game between themselves.

Let V^{ii}, V^{ij} denote the altruistic utilities of a parent of cultural trait i for a child of trait i, j, respectively. Parents display *imperfect empathy*, that is, they evaluate their children's utility with their own (the parents') preferences.[8] Parents are altruistic toward their children and grand-children but not toward the offspring of all other future generations.[9] Furthermore, at each time t, parental socialization entails a direct socialization cost $C(\tau^i)$. Parents, however, do not internalize their children and grand-children direct socialization costs.[10] These assumptions - plus some (technical) regularity - can be formally reduced to the following.

Assumption 1. *Parental utility with respect to the first generation (t) and their expected utility*

[8] This form of imperfect altruism - in Bisin and Verdier (2001) - allows for parental socialization choices which are not necessarily aligned with their own cultural trait in more complex socio-economic interactions.

[9] Parents are therefore *time-inconsistent hyperbolic discounters over generations*. The canonical model in the literature - e.g., Phelps and Pollak (1968) and Bisin and Verdier (2001) - has parents which are altruistic only towards their children. Adding grand-children is natural in that parents can expect to have direct personal interactions with children and grand-children in their lifetime. It can be shown that, generally, time-inconsistency will tend to reduce parental socialization choices at equilibrium; see Proposition A.1 and A.2 in Appendix A.

[10] This assumption can be justified as a consequence of imperfect empathy, whereby parents only care about children and grand-children when young. It greatly simplifies the analysis.

with respect to the second generation $(t+1)$ are, respectively:

$$U_t^i = P_t^{ii}\Delta V^i + V^{ij} - C(\tau_t^i), \tag{3}$$
$$U_{t+1}^i = P_t^{ii}[P_{t+1}^{ii}\Delta V^i + V^{ij}] + (1-P_t^{ii})[P_{t+1}^{ji}\Delta V^i + V^{ij}]$$

where $\Delta V^i = V^{ii} - V^{ij} > 0$ and $C(\tau)$ is strictly increasing and convex.

Utility U_{t+1}^i is the weighted average of the utilities realized when the child is of trait i or of trait j at t. The transition probabilities depend on socialization choices as well as on the distribution of the population, by Equation (1):

$$P_t^{ii} = P_t^{ii}(\tau_t^i, q_t^i); \;\; P_{t+1}^{ii} = P_{t+1}^{ii}(\tau_{t+1}^i, q_{t+1}^i); \quad P_{t+1}^{ji} = P_{t+1}^{ji}(\tau_{t+1}^j, 1-q_{t+1}^i).$$

3 Equilibrium

In this section we study parental socialization and compare it with the socialization of cultural leaders at equilibrium. We assume that the socialization problem at time t is constructed by anticipating a one-period ahead socialization problem at $t+1$ and no socialization problem at $t+2$. This is just for simplicity and essentially without loss of generality. In Appendix B we define and extend the analysis to the Markovian problem where every agent - parent or cultural leader - anticipates two-period ahead socialization problems.

3.1 Parental socialization

Parental socialization choice at time t takes as given q_t^i (which is pre-determined) as well as q_{t+1}^i.

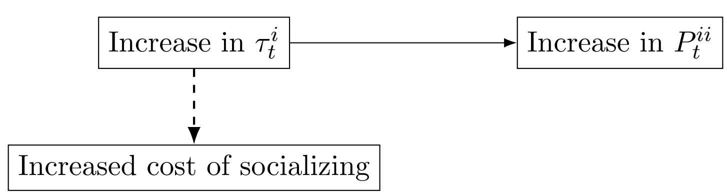


Figure 1: Effects of an increase of τ_t^i for a myopic parent.

Figure (1) summarizes the different positive (solid lines) and negative (dashed lines) effects that increasing socialization at t has on the problem of parents of cultural trait i from the point of view of a myopic parent.

Definition 3.1. Parental socialization equilibrium at t, given $q_t^i \in [0,1]$, is $(\tau_t^i, \tau_t^j) \in [0,1]^2$ and

$q^i_{t+1} \in [0,1]$ such that, for each $i \in \{a,b\}$, letting $j \neq i$,

$$\begin{aligned} &\tau^i_t \in \arg\max U^i_t + U^i_{t+1} \\ &\text{subject to: } P^{ii}_t = \tau^i_t + (1-\tau^i_t)q^i_t, \qquad P^{ij}_t = (1-\tau^i_t)(1-q^i_t), \\ &\qquad\qquad\quad \tau^i_{t+1} = (1-q^i_{t+1})\Delta V^i, \qquad \tau^j_{t+1} = q^i_{t+1}\Delta V^j, \\ &\qquad\qquad\quad q^i_{t+1} \text{ given,} \end{aligned} \tag{4}$$

and

$$q^i_{t+1} - q^i_t = q^i_t(1-q^i_t)(\tau^i_t - \tau^j_t). \tag{5}$$

3.1.1 Cultural leaders' socialization

Consider an environment where the socialization choices are taken by a cultural leader. The leader controls parental socialization at time t. He/she understands the effect that changes in socialization have for the future generations and evaluates the preferences of parents of group i with $\delta = 1$. The leader's socialization choice at time t takes as given q^i_t (which is pre-determined) but anticipates the effects τ^i_{t+1}, τ^j_{t+1}, q^i_{t+1}, which are determined at equilibrium. Leaders at t anticipate the choice of $t+1$ leaders; that is, they are choosing τ^i_{t+1}, τ^j_{t+1} to maximize one-period utility.[11] Specifically, at $t+1$ the anticipated choices are $\tau^i_{t+1} = (1-q^i_{t+1})\Delta V^i$ and $\tau^j_{t+1} = q^i_{t+1}\Delta V^j$.

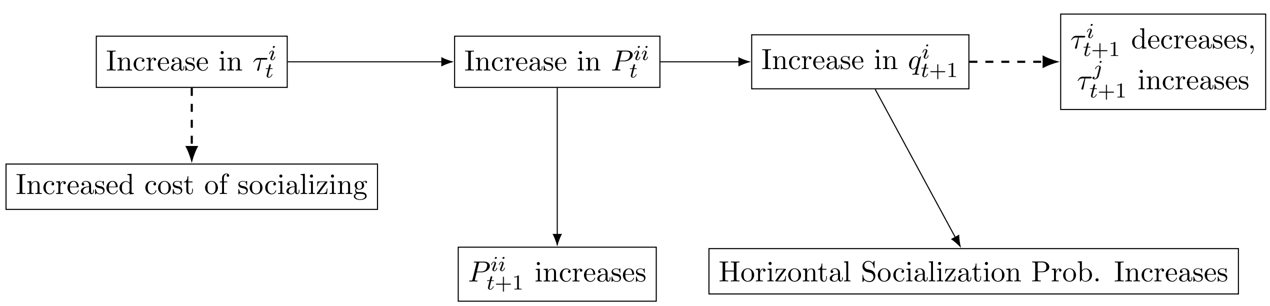


Figure 2: Effects of an increase of τ^i_t.

Figure (2) summarizes the different positive (solid lines) and negative (dashed lines) effects that increasing socialization at t has on the socialization problem of culture i for forward-looking agents. By our definition of equilibrium parents will ignore all the effects related to the increase of q^i_{t+1} (right side of the graph), while leaders take all the effects into account.

Leaders, therefore, play a strategic game with each other. We consider Nash Equilibrium as the

[11] Equivalently, it could be parents choosing τ^i_{t+1}, τ^j_{t+1} to maximize one-period utility.

equilibrium concept of the game.[12]

Definition 3.2. A cultural leaders' socialization equilibrium at t, given $q_t^i \in [0,1]$, is $(\tau_t^i, \tau_t^j) \in [0,1]^2$ and $q_{t+1}^i \in [0,1]$ such that, for each $i \in \{a,b\}$, letting $j \neq i$,

$$\begin{aligned} &\tau_t^i \in \arg\max U_t^i + U_{t+1}^i \qquad (6)\\ &\text{subject to: } P_t^{ii} = \tau_t^i + (1-\tau_t^i)q_t^i, \qquad P_t^{ij} = (1-\tau_t^i)(1-q_t^i),\\ &\qquad \tau_{t+1}^i = (1-q_{t+1}^i)\Delta V^i, \qquad \tau_{t+1}^j = q_{t+1}^i \Delta V^j,\\ &\qquad q_{t+1}^i - q_t^i = q_t^i(1-q_t^i)(\tau_t^i - \tau_t^j),\\ &\qquad \tau_t^j \text{ given.} \end{aligned}$$

3.1.2 Comparison between parental and cultural leaders' socialization

The main difference between the socialization choice of parents and cultural leaders consists in the fact that the leaders internalize the effect of their own choice on future socialization via social interactions; while parents do not. Under a relatively weak sufficient condition, this effect is positive: a higher socialization effort increases the future population share of the trait and hence future socialization via social interactions to that trait, other things equal. As a consequence, cultural leaders will tend to choose a higher socialization effort than parents.

Proposition 3.1. *Under the sufficient conditions that,*

$$q_{t+1}^i \, \Delta V^j < \frac{1}{2} \text{ and } (1-q_{t+1}^i)\, \Delta V^i < \frac{1}{2}, \quad \text{for all } q_t^i \in (0,1), \tag{7}$$

both the cultural leaders of group i and j choose higher socialization effort than the respective parents.

We sketch the proof here, while the details are in Appendix A. Consider socialization of group i. The leader internalizes the effect of its choice τ_t^i on q_{t+1}^i; while parents do not. It is useful then to decompose the effect of τ_t^i on q_{t+1}^i to better understand the difference between the parental and the leaders' choices of socialization at t. From the FOC's - reported in the proof of Proposition A.3 in Appendix A, this effect takes the following form:

$$q_t^i(1-q_t^i)\Delta V^i \Big(P_t^{ii}\big(1 - 2\Delta V^i + 2q_{t+1}^i \Delta V^i\big) + (1-P_t^{ii})\big(1 - 2q_{t+1}^i \Delta V^j\big)\Big) \tag{8}$$

[12] Existence is straightforward; see Appendix A.

Intuitively, there is a direct positive effect of socializing more for the leader of trait i, in that the increase in q^i_{t+1} he/she can induce by increasing τ^i_t increases oblique socialization to trait i for the generation of grandchildren - independently of the trait of the generation of children. But there is also an indirect effect which goes in the opposite direction. An increase in q^i_{t+1} induces the generation of children of trait i to reduce their direct socialization choice and the generation of children of trait j to increase it. In the following we construct sufficient conditions for the total effect - represented by Equation (8) - to be positive. We then show that the sufficient condition in the statement guarantees that (8) is positive.

While not straightforward to study in closed form, the sufficient conditions in Proposition 3.1 are shown to be quite tight in our simulations, as summarized by Figure 3.[13] There are two different regions where the sufficient condition does not hold, colored in orange and pink.[14] Consider group i for illustration. The leader socializes less than the parents if i) ΔV^i is high and ΔV^j is low; ii) q^i_t is low. A high ΔV^i implies that the culture i at $t+1$ will respond aggressively to changes in q^i_{t+1} and decrease socialization by a lot, especially when starting from a low q^i_t; in turn a low ΔV^j implies that group j will not socialize too much, especially when starting from a high q^j_t (that is, a low q^i_t). Under these conditions then it is optimal for the leader of group i to keep the population fraction of its group next period, q^i_{t+1}, lower relatively to the population fraction that would be induced by parental choice.

4 Welfare Analysis

We adopt a notion of social welfare constructed on the ex-ante preferences of the parents, to pursue our welfare analysis of socialization choice.

Definition 4.1. Efficient socialization at t, given q^i_t, is the set of $(\tau^i_t, \tau^j_t) \in [0,1]^2$ and $q^i_{t+1} = 1 - q^j_{t+1}$ such that - for some Pareto weight $0 \leq \alpha \leq 1$:

[13] All simulations in the two-period case rely on computing the fixed point of the joint system of first order conditions for the parents and the leaders respectively.

[14] If these graphs were generated by taking $\Delta V^i = 0.25 + \beta$, $\Delta V^j = 0.25 - \beta$ and varying $\beta \in [-0.25, 0.25]$, cultural leaders would always socialize more than parents.

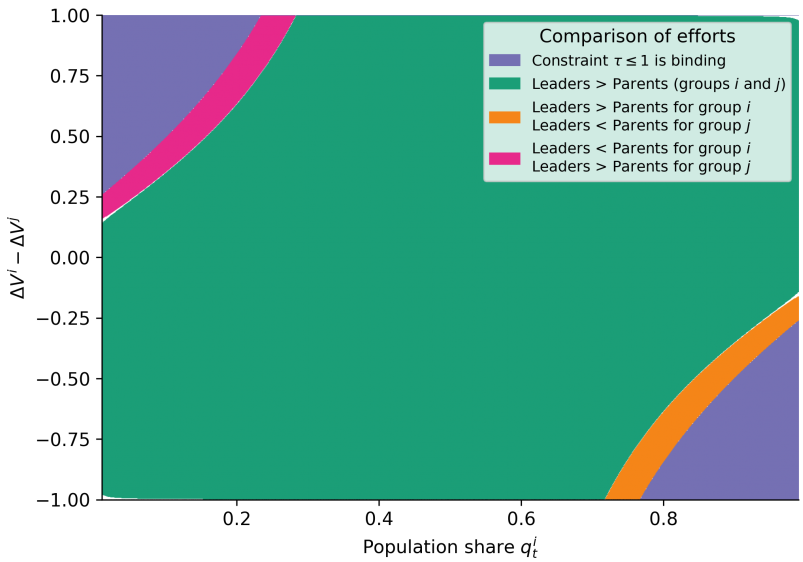


Figure 3: Qualitative comparison of equilibrium socialization effort under parental and cultural-leader socialization.

$$(\tau_t^i, \tau_t^j) \in \arg\max \alpha(U_t^i + U_{t+1}^i) + (1-\alpha)(U_t^j + U_{t+1}^j) \tag{9}$$

$$\begin{aligned} \text{subject to: } & P_t^{ii} = \tau_t^i + (1-\tau_t^i)q_t^i; \quad P_t^{ij} = (1-\tau_t^i)(1-q_t^i) \\ & P_t^{jj} = \tau_t^j + (1-\tau_t^j)(1-q_t^i); \quad P_t^{ji} = (1-\tau_t^j)q_t^i \\ & \tau_{t+1}^i = (1-q_{t+1}^i)\Delta V^i; \quad \tau_{t+1}^j = q_{t+1}^i \Delta V^j \\ & q_{t+1}^i - q_t^i = q_t^i(1-q_t^i)(\tau_t^i - \tau_t^j) \end{aligned}$$

This notion allows to study the welfare properties of cultural interaction between groups without specifying pre-determined roles - like immigrants and natives - or sizes - like majoritarian and minoritarian.

We first obtain the following result.

Proposition 4.1. *Under the sufficient condition (7) parental socialization choice is generically efficient.*

This result - proved in Appendix A - is somewhat counter-intuitive in that efficiency is obtained even though at equilibrium parents do not internalize their effect on q_{t+1}^i when choosing socialization. This is indeed a sort of externality: a perturbation in τ_t^i has an effect on q_{t+1}^i which in turn has effects of different sign on the utility of two groups which are not necessarily symmetric in size. The effects of τ_t^i and τ_t^j on q_{t+1}^i are however symmetric: the law of motion of q_t^i depends on $\tau_t^i - \tau_t^j$. A utility transfer therefore cannot be obtained by a perturbation of τ_t^i and a corresponding counter-perturbation in τ_t^j.[15]

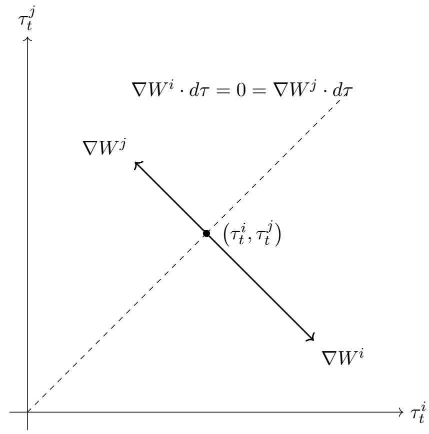


Figure 4: Gradient vectors and improvement half-spaces at the parental equilibrium.

Formally, this symmetry is represented in Figure 4. At the parental optimal choice (τ_t^i, τ_t^j) the two groups' utility gradient vectors, evaluated at their parental equilibrium choice, are linearly dependent and have opposite direction. The two strict orthogonal half-spaces to the gradient vectors - that is, the socialization choices that represent welfare improvements for each group - do not intersect and therefore no perturbation of (τ_t^i, τ_t^j) can induce a social welfare improving deviation.

While transfers across groups cannot be implemented simply via perturbation of effort choices at the parental equilibrium, we may be able to instead construct social welfare improving deviations by enlarging the control space in the social welfare function maximization. This is indeed the case. More specifically, assume that now we allow for transfers across groups, via socialization costs. That is, for each of τ_t^i, τ_t^j, we allow social welfare to be maximized also with respect to the fraction of the cost that group i pays for the socialization of group j, and vice versa. Let this fraction be denoted θ^{ij}. The utility of group i in the first time period is going to be $U_t^i = P_t^{ii}\Delta V^i + V^{ij} - (1-\theta^{ji})C(\tau_t^i) - \theta^{ij}C(\tau_t^j)$.

Proposition 4.2. *Allowing for socialization cost transfers across agents at social welfare,* $0 \leq \theta^{ij}, \theta^{ji} \leq 1$, *parental socialization is inefficient generically for* $(q_t^i, \Delta V^i, \Delta V^j) \in [0,1]^3$.

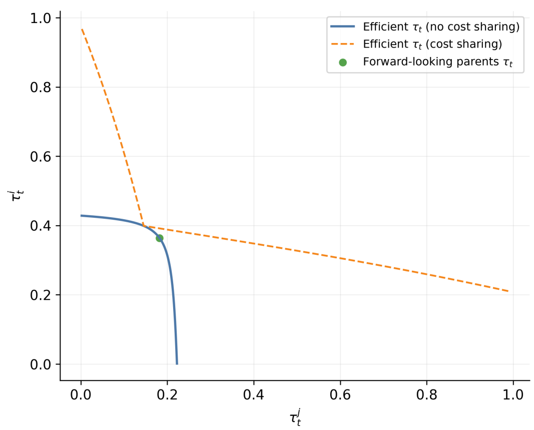
The proof - in Appendix A - follows straightforwardly from the existence of the externality.

[15] This is the case when both cultural leaders choose higher socialization effort than the respective parents - hence the role of condition (7).

Figure 5a shows how the optimal socialization choices of the planner change when we allow for cost transfers between groups. The leaders' choice is still above the optimal, while parents under-socialize relative to the planner's choice.

At the social welfare optimum group j pays all the cost whenever $\alpha > \frac{1}{2}$ and group i pays all the cost whenever $\alpha < \frac{1}{2}$. Figure 5b shows the optimal curves - for the two groups - with $\alpha \in [0,1]$. The efficient choice frontier in Figure 5a is the pointwise maximum of these two curves, and their intersection is the point where $\alpha = \frac{1}{2}$.

(a) Planner's Pareto socialization choices with endogenous cost sharing.



(b) Efficient socialization choices under the extreme cases in which one group bears the entire socialization cost of both groups.

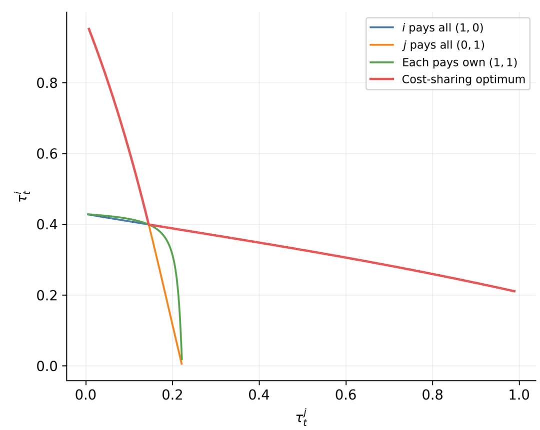


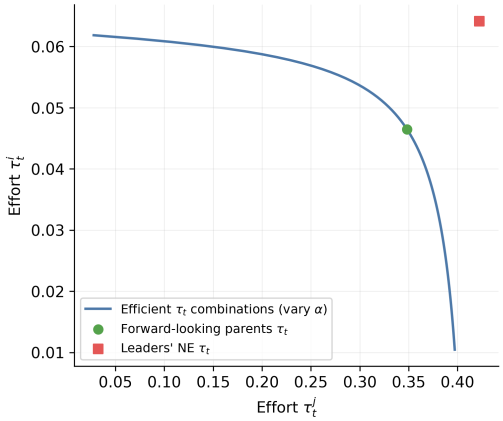
Figure 5: Planner outcomes under alternative welfare specifications. The left panel allows the planner both to choose socialization efforts and to redistribute socialization costs across groups; the right panel isolates the benchmark full-cost allocations that determine the efficient frontier.

We can also study the welfare properties of the Nash equilibrium between cultural leaders.

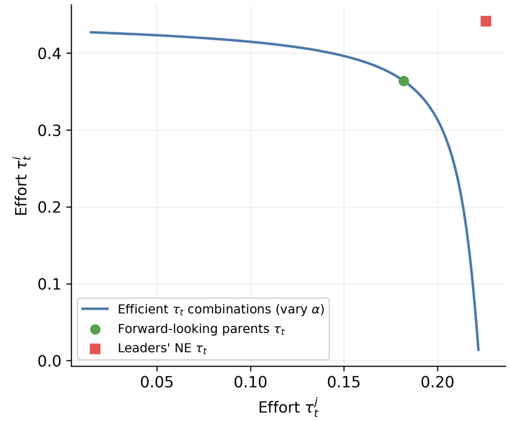
Proposition 4.3. *Nash Equilibrium choices of cultural leaders at time t for both group i and j are generically Pareto inefficient. Further, if $q^i_{t+1}\,\Delta V^j < \frac{1}{2}$ and $(1 - q^i_{t+1})\,\Delta V^i < \frac{1}{2}$ leaders socialize too much relative to the optimal choice.*

This result captures a fundamental inefficiency due to strategic complementarity in cultural transmission - the proof in Appendix A makes this apparent. Since cultural leaders from both cultural groups take into account the effects of their socialization choices on their next generations, a Pareto improvement with respect to their Nash equilibrium choices can be obtained by symmetrically reducing their effort with respect to their choices at equilibrium. This can be done so as to maintain their utility from socialization constant (while gaining from the reduction in socialization

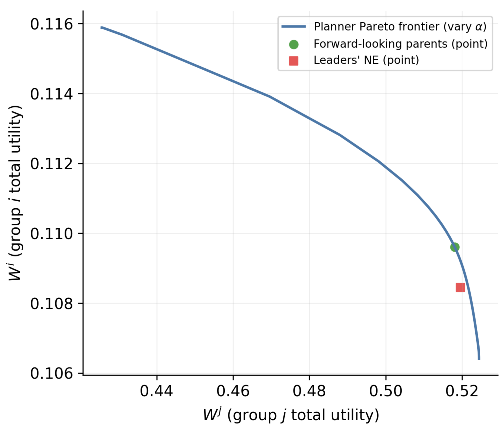
(a) Planner choices for $q_t^i = 0.60$ and $(\Delta V^i, \Delta V^j) = (0.10, 0.50)$.



(b) Planner choices for $q_t^i = 0.50$ and $(\Delta V^i, \Delta V^j) = (0.60, 0.30)$.



(c) Pareto frontier for $q_t^i = 0.60$ and $(\Delta V^i, \Delta V^j) = (0.10, 0.50)$.



(d) Pareto frontier for $q_t^i = 0.50$ and $(\Delta V^i, \Delta V^j) = (0.60, 0.30)$.

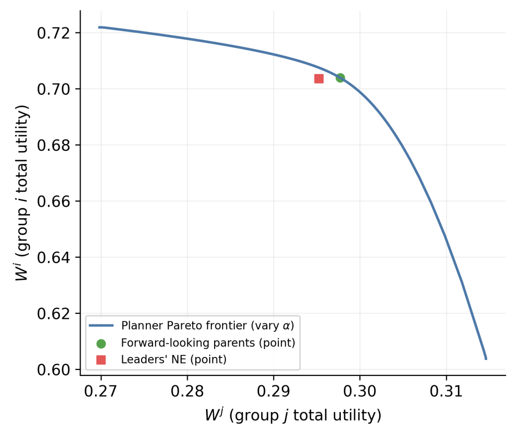


Figure 6: The top row reports the socialization choices (τ_t^i, τ_t^j) selected by the planner as the Pareto weight varies. The bottom row reports the associated utility frontiers for the same parameter values.

costs).[16]

When the inefficiency is symmetric the actual path of q_t^i is unaffected: the excess socialization that strategic complementarities generate is pure waste, sustaining the same cultural trajectory at needlessly higher cost rather than pushing assimilation or differentiation forward at an inefficient pace. Distortion of the evolutionary path itself instead arises from asymmetries: whenever one group's institutions systematically over- or under-socialize relative to the other's, the population share q_t^i evolves too quickly or too slowly relative to what a planner weighing both groups' ex-ante

[16] In Bisin and Verdier (2001) - Proposition 3, p. 311-12 - it is claimed that, a similar logic implies that the parental socialization choices (τ_t^i, τ_t^j) are inefficient even in the case in which parents only care about their children. This is incorrect. While it is true that parental choices (τ_t^i, τ_t^j) are not cost effective for a given dynamic path of the population, parental preferences do not depend on future population dynamics in this case.

preferences would choose. In this sense, the paper's welfare results speak less to whether cultural evolution proceeds "too fast" or "too slow" in general, and more to when institutional mediation — parental or by cultural leaders — introduces a wedge between groups' relative efforts, and when it merely raises the cost of sustaining an otherwise undistorted cultural trajectory.

5 Conclusions

We have studied the efficiency properties of cultural transmission processes governed by parental socialization choices or by cultural leaders. Our main results are that both socialization processes are rather generally inefficient. In particular, intuitively, parental socialization is inefficient because the group that would benefit most from increased effort has no mechanism to compensate the other group for the negative externalities this would create; cultural leaders' socialization is instead inefficient due to the strategic complementarities in their choice problems. As a result parents socialize less compared to the optimal, while cultural leaders socialize too much.

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A Additional Propositions and Proofs

Proposition A.1. *The socialization choice of parent i at time t, τ_t^i is larger when the parent cares about children and grand-children than when he/she only cares about children.*

Proof. Consider first the case in which parents only care about children. In this case, a parent of trait i chooses $\tau^i \in \mathbb{R}_+^n, x_t^i \in X$ to maximize:

$$\max_{x^i,\tau^i} u^i(x^i) + P_t^{ii}V^{ii} + P_t^{ij}V^{ij} - C(\tau^i) \tag{10}$$

Restricting $C(\tau^i)$ to be quadratic - $C(\tau^i) = \frac{1}{2}(\tau^i)^2$, with relative loss of generality - we have

$$\tau_t^i = (1 - q_t^i)\Delta V^i,$$

where $\Delta V^i = V^{ii} - V^{ij}$ represents the *cultural intolerance* of cultural group i.[17]

When instead parents care about their children and grand-children, their socialization choice at time t affects U_{t+1}^i through its effect on P_t^{ii}. In fact, τ_t^i will be larger than with myopic parents (that is, larger than $(1 - q_t^i)\Delta V^i$) as long as, $[P_{t+1}^{ii}\Delta V^i + V^{ij}] > [P_{t+1}^{ji}\Delta V^i + V^{ij}]$; that is, when

$$P_{t+1}^{ii} > P_{t+1}^{ji}.$$

But indeed - for any $\tau_{t+1}^i > 0$ and/or $\tau_{t+1}^j < 1$, it is the case that $P_{t+1}^{ii} > q_{t+1}^i$ while $P_{t+1}^{ji} < q_{t+1}^i$. $\square$

Note that the result does not depend on the equilibrium determination of expectations. We only assume that $\tau_t^i \in (0, 1)$, a simple regularity assumption which will generally hold at equilibrium. A more general result, showing that less-time-inconsistent (more forward-looking) parents socialize more, is stated and proved next.

Let the general utility of parents of type i be written as follows:

$$\begin{aligned} U_t^i &= \delta(P_t^{ii}V^{ii} + P_t^{ij}V^{ij}) + \alpha_1\delta^2(P_{t,t+1}^{ii}V^{ii} + P_{t,t+1}^{ij}V^{ij}) + \dots \\ &= \underbrace{\delta(P_t^{ii}V^{ii} + P_t^{ij}V^{ij})}_{\text{altruistic utility to parents from children}} + \underbrace{\sum_{h=1}^{\infty} \alpha_h\delta^{h+1}(P_{t,t+h}^{ii}V^{ii} + P_{t,t+h}^{ij}V^{ij})}_{\text{altruistic utility to parents from succeeding generations.}} \end{aligned}$$

Proposition A.2. *Let parent i discount future generations according to $\delta = 1$ and the sequence*

[17] Direct socialization is naturally interpreted as a probability - and hence V^{ii}, V^{ij} need to be appropriately restricted so that the solution of problem (10) is a map $\tau_t^i : [0, 1] \to [0, 1]$. With quadratic costs, we require $0 \le \Delta V^i \le 1$.

$\{a_h\}_{h\geq 0}$ *given by*

$$a_h = \begin{cases} 1, & h < k, \\ 0, & h \geq k \end{cases}$$

Then the optimal socialization effort increases in k*.*

Proof. Consider the problem at $t = 0$ without loss of generality. Let $P^{ii}_{0,t}(\tau^i_0)$ be the probability that a parent of type i has a **t+1**-th generation descendant of type i, as a function of the initial socialization choice τ^i_0. The expected utility of parents of type i at time zero that look forward until generation h is:

$$U^h(\tau^i_0) = \sum_j^h \left[P^{ii}_{0,t}(\tau^i_0)\Delta V^i + V^{ij}\right] - C(\tau^i_0)$$

Taking the FOC, the optimal socialization choice for the parent is:

$$\tau^i_t = \sum_{t=0}^{h} \frac{\partial}{\partial \tau^i_t} P^{ii}_{0,t}(\tau^i_0)\Delta V^i$$

It suffices to show the case where parents with $h = k$ socialize more than parents with $h = k-1$. The general result follows immediately. It therefore suffices to show that:

$$\sum_{t=0}^{k} \frac{\partial}{\partial \tau^i_0} P^{ii}_{0,t}(\tau^i_0)\Delta V^i - \sum_{t=0}^{k-1} \frac{\partial}{\partial \tau^i_0} P^{ii}_{0,t}(\tau^i_0)\Delta V^i = \frac{\partial}{\partial \tau^i_0} P^{ii}_{0,k}(\tau^i_0)\Delta V^i > 0$$

We proceed by induction on k. The case for $k = 1$ is given by Proposition *A.1*. Ignoring the constant ΔV^i, assume that $\frac{\partial}{\partial \tau^i_0} P^{ii}_{0,k-1}(\tau^i_0) > 0$. We have:

$$\frac{\partial}{\partial \tau^i_0} P^{ii}_{0,k}(\tau^i_0) = \frac{\partial}{\partial \tau^i_0}\left[P^{ii}_{0,k-1}P^{ii}_{k-1,k}(\tau^i_0) - (1 - P^{ii}_{0,k-1})P^{ji}_{k-1,k}\right] = \left[P^{ii}_{k-1,k} - P^{ji}_{k-1,k}\right]\frac{\partial}{\partial \tau^i_0} P^{ii}_{0,k-1}(\tau^i_t)$$

Since $P^{ii}_{k-1,k} > P^{ji}_{k-1,k}$ for all possible choices of τ^i_k, τ^j_k the last term is always positive. □

Proposition A.3. *A Cultural leaders' socialization equilibrium - in Definition 3.2 - exists.*

Proof. From the first order conditions of the maximization problems in the cultural leaders' problem, it is straightforward to show that the Nash Equilibrium is a set of strategies (τ^i_t, τ^j_t) that simultaneously satisfies:

$$\tau^i_t = (1 - q^i_t)\,\Delta V^i \left[1 + \left(P^{ii}_{t+1} - P^{ji}_{t+1}\right) + q^i_t\Big(P^{ii}_t\big(1 - 2\Delta V^i + 2q^i_{t+1}\Delta V^i\big) + (1 - P^{ii}_t)\big(1 - 2q^i_{t+1}\Delta V^j\big)\Big)\right]$$

$$\tau^j_t = (1 - q^j_t)\,\Delta V^j \left[1 + \left(P^{jj}_{t+1} - P^{ij}_{t+1}\right) + q^j_t\Big(P^{jj}_t\big(1 - 2\Delta V^j + 2q^j_{t+1}\Delta V^j\big) + (1 - P^{jj}_t)\big(1 - 2q^j_{t+1}\Delta V^i\big)\Big)\right]$$

where q^i_{t+1} and $q^j_{t+1} = 1 - q^i_{t+1}$ depend on τ^i_t and τ^j_t. A Nash equilibrium exists by a straightforward consequence of Brouwer Fixed Point Theorem. □

Proof of Proposition 3.1. We first prove the following lemma.

Lemma A.1. *A sufficient condition guaranteeing for all* $q^i_t \in (0,1)$ *is the following:*

$$q^i_{t+1}\,\Delta V^j < \frac{1}{2} \text{ and } (1 - q^i_{t+1})\,\Delta V^i < \frac{1}{2}.$$

A simpler (but stricter) sufficient condition on cultural intolerance levels directly is: $\max\{\Delta V^i, \Delta V^j\} < \frac{1}{2}$.

Proof. Define the affine function

$$A(P^{ii}_t) \;=\; \big(1 - 2q^i_{t+1}\Delta V^j\big) \;+\; 2P^{ii}_t\big[q^i_{t+1}(\Delta V^j + \Delta V^i) - \Delta V^i\big], \qquad 0 \le P^{ii}_t \le 1.$$

Because $A(P^{ii}_t)$ is linear in P^{ii}_t, its minimum on the compact set $[0,1]$ is attained at one of the endpoints:

$$A(t) = 1 - 2q^i_{t+1}\Delta V^j, \qquad A(t+1) = 1 - 2(1 - q^i_{t+1})\Delta V^i.$$

The two inequalities in the statement ensure $A(t) > 0$ and $A(t+1) > 0$; hence $A(P^{ii}_t) \ge \min\{A(t), A(t+1)\} > 0$ for every $P^{ii}_t \in [0,1]$.

As for the condition on cultural intolerance, $\max\{\Delta V^i, \Delta V^j\} < \frac{1}{2}$, with $q^i_{t+1} \in (0,1)$ directly implies $q^i_{t+1}\Delta V^j < \frac{1}{2}$ and $(1 - q^i_{t+1})\Delta V^i < \frac{1}{2}$. □

Note that these conditions - though referring to the effects of τ^i_t on q^i_{t+1} - are symmetric and hence they also guarantee that the effect of τ^j_t on $q^j_{t+1} = 1 - q^i_{t+1}$ is positive. These conditions however do not guarantee that the socialization choices of cultural leaders i and j at t are higher than those of the respective parents.[18] Formally, both the first order conditions of the leaders and the parents contain a term $P^{ii}_{t+1} - P^{ji}_{t+1}$, for group i - and $P^{jj}_{t+1} - P^{ij}_{t+1}$, for group j - which are evaluated however at different values of q^i_{t+1} by the parents and the leaders.

We can now proceed to prove the statement of the proposition. Note that the objective functions are strictly concave, and thus there exists a unique maximizer. Let $\frac{\partial U^L_i}{\partial \tau^i_t}$ and $\frac{\partial U^i_P}{\partial \tau^i_t}$ denote the first order condition of the leaders and parents of group i respectively. Notice that $\frac{\partial U^i_L}{\partial \tau^i_t} = \frac{\partial U^i_P}{\partial \tau^i_t} + q^i_t(1 - q^i_t)\Delta V^i A(P^{ii}_t)$, where A is defined as in Lemma A.1. Denote the parental equilibrium choices at t by $\tau_P = (\tau^i_P, \tau^j_P)$. The first order condition of the parental problem evaluated at the equilibrium

[18] Our simulations however suggest that this is indeed generally the case, except perhaps in extreme cases.

choices imply that $\left.\frac{\partial U_P^i}{\partial \tau_t^i}\right|_{\tau_P} = 0$ and therefore $\left.\frac{\partial U_L^i}{\partial \tau_t^i}\right|_{\tau_P} = q_t^i(1-q_t^i)\Delta V^i A(P_t^{ii})$ which is positive by our assumption. Therefore the equilibrium choice of τ_t^i for the leaders must be strictly higher than the parents. An identical argument applies to group j. □

Proof of Proposition 4.1. The proof is straightforwardly exploiting a subtle symmetry.

Proof. Let $W^i((\tau_t^i)^*)$ and $W^j((\tau_t^j)^*)$ denote the values, for i and j respectively, of problem 9. Evaluating the partials of $W^i((\tau_t^i)^*)$ and $W^j((\tau_t^j)^*)$ at the optimal choice, (including the effect of socialization on q_{t+1}^i which parents ignore):

$$\nabla W^i((\tau_t^i)^*) = \begin{pmatrix} q_t^i(1-q_t^i)\Delta V^i A(P_t^{ii}) \\ \\ -q_t^i(1-q_t^i)\Delta V^i A(P_t^{ii}) \end{pmatrix} \text{ and } \nabla W^j((\tau_t^j)^*) = \begin{pmatrix} -q_t^i(1-q_t^i)\Delta V^j B(P_t^{jj}) \\ \\ q_t^i(1-q_t^i)\Delta V^j B(P_t^{jj}) \end{pmatrix}$$

where $A(P_t^{ii}) = \left(1-2q_{t+1}^i\Delta V^j\right) + 2P_t^{ii}\left[q_{t+1}^i(\Delta V^j+\Delta V^i)-\Delta V^i\right]$ as in Proposition A.1 and $B(P_t^{jj}) = \left(1-2\Delta V^i+2q_{t+1}^i\Delta V^i\right) + 2P_t^{jj}\left[(1-q_{t+1}^i)(\Delta V^i+\Delta V^j)-\Delta V^j\right]$. Assuming that A and B have the same sign, any symmetric local perturbation thus leaves utility unchanged for both parents. But the terms A, B are, respectively, the effects of τ_t^i, τ_t^j on $q_{t+1}^i, q_{t+1}^j = 1-q_{t+1}^i$ in the first order conditions of the leaders' problem (see Proposition A.1) and therefore the assumptions made guarantee that they are both positive.[19] □

Proof of Proposition 4.2. The proof follows straightforwardly from the existence of the externality.

Proof. Allowing the planner to control the distribution of the costs, there is a Pareto improving deviation. Let $a := q_t^i(1-q_t^i)\Delta V^i A(P_t^{ii}), b := q_t^i(1-q_t^i)\Delta V^j B(P_t^{jj})$ (i.e., the terms from the planner's first order conditions evaluated at the parental choice). (It is possible to show that) In a zero-Lebesgue set of $(q_t^i, \Delta V^i, \Delta V^j) \in [0,1]^3$, it is the case that $a \neq b$. Consider then, without loss of generality, the case in which $a > b$; that is, group i benefits more from an increase in socialization than group j. Consider the perturbation $d\tau_t^i = \epsilon$, $d\tau_t^j = 0$ where $\epsilon > 0$. The change in utility for the two groups is: $dW^i = a\epsilon$ and $dW^j = -b\epsilon$ respectively. Now take the perturbation $d\theta^{jj} = -\frac{2b}{(\tau_t^j)^2}\epsilon$.

[19] In principle it could be that A and B have opposite signs. For example in the case where $A(P_t^{ii}) > 0$ and $B(P_t^{jj}) < 0$ a local positive perturbation of τ_t^i is positive for both groups. However it is difficult to say when this happens, and in our simulations in almost all cases the parents ended up on the Pareto frontier. See Figure 6: panels c and d report some graphs of the Pareto frontier from the simulations, while panels a and b report the choices of (τ_t^i, τ_t^j) for the planner, leaders and parents in the case where the planner has no control over the costs.

This change[20] offsets the change in utility for group j since $dW^j = -b\epsilon + \frac{1}{2}(\tau_t^j)^2 \frac{2b}{(\tau_t^j)^2}\epsilon = 0$. Now for group i the total change in utility is $dW^i = a\epsilon - \frac{1}{2}(\tau_t^j)^2 \frac{2b}{(\tau_t^j)^2}\epsilon = \epsilon(a-b) > 0$. □

Proof of Proposition 4.3. The proof follows straightforwardly from the strategic complementarity embedded in the cultural leaders' game.

Proof. Suppose that at the Nash Equilibrium, we operate the infinitesimally small perturbation

$$|d\tau_t^i| = |d\tau_t^j| = \epsilon.$$

The change of utility in the parents is:

$$\nabla W^i(\tau_t^i)^* \cdot d\tau \text{ and } \nabla W^j(\tau_t^i)^* \cdot d\tau$$

Since the choices of the leaders are optimal, we have:

$$\frac{\partial}{\partial \tau_t^i} W^i((\tau_t^i)^*) = 0 \text{ and } \frac{\partial}{\partial \tau_t^j} W^j((\tau_t^j)^*) = 0$$

so it remains to verify that $\frac{\partial}{\partial \tau_t^j} W^i((\tau_t^i)^*) \neq 0$. Calculating the partial derivative we get:

$$\frac{\partial W^i}{\partial \tau_t^j} = -\, q_t^i\big(1-q_t^i\big)\, \Delta V^i \underbrace{\Big[P_t^{ii}\big(1-2\,(1-q_{t+1}^i)\,\Delta V^i\big) + (1-P_t^{ii})\big(1-2\,q_{t+1}^i\,\Delta V^j\big)\Big]}_{=A(P_t^{ii})}. \tag{11}$$

Where $A(P_t^{ii})$ is the same as in Lemma A.1. Call $B(P_t^{jj})$ the respective term for group j. The deviation:

$$d\tau_t^i = -\text{sign}(B(P_t^{jj}))\epsilon,\ d\tau_t^j = -\text{sign}(A(P_t^{ii}))\epsilon$$

is Pareto improving. If the conditions of the second part of the propositions hold, then $\text{sign}(A(P_t^{ii})) = \text{sign}(B(P_t^{jj})) > 0$ so the Pareto improving perturbation is to decrease effort for both groups. □

[20] Observe that $\partial_{\theta^{jj}} W^j = -\frac{1}{2}(\tau_t^j)^2$, $\partial_{\theta^{jj}} W^i = \frac{1}{2}(\tau_t^j)^2$.

B Infinite-Horizon Markov Formulation

We extend the socialization problem to one with infinitely many generations, each of which only cares about the next two generations. Preferences at time t are the same as in the two-period case of Section 3, but now parents/leaders anticipate that choices at time $t+1$ will be made by another parent/leader with the same two-period preferences.

B.1 Parental Equilibrium

Definition B.1. Parental socialization equilibrium at time t, given $q_t^i \in [0,1]$, is $(\tau_t^i, \tau_t^j) \in [0,1]^2$ and $q_{t+1}^i \in [0,1]$ such that, for each $i \in \{a,b\}$, letting $j \neq i$,

$$
\begin{aligned}
\tau_t^i \in \arg\max\ & U_t^i + U_{t+1}^i \qquad (12)\\
\text{subject to:}\quad & P_t^{ii} = \tau_t^i + (1-\tau_t^i)q_t^i, \qquad P_t^{ij} = (1-\tau_t^i)(1-q_t^i),\\
& (\tau_{t+1}^i, \tau_{t+1}^j) \text{ satisfy the same equilibrium condition at time } t+1 \text{ given } q_{t+1}^i,\\
& q_{t+1}^i \text{ given},
\end{aligned}
$$

and

$$q_{t+1}^i - q_t^i = q_t^i(1-q_t^i)(\tau_t^i - \tau_t^j). \qquad (13)$$

Let $\tau_P^i(q_t^i)$ denote the optimal parental socialization of group i when the share of trait i in the population is q_t^i, which does not depend on the specific time t. As in the two-period case, the first order condition of parents of type i at time t can be written as

$$\tau_P^i(q_t^i) = (1-q_t^i)\Delta V^i\big(1 + P_{t+1}^{ii} - P_{t+1}^{ji}\big),$$

where P_{t+1}^{ii} and P_{t+1}^{ji} are computed using the same policy functions that parents use at time t:

$$P_{t+1}^{ii} = \tau_P^i(q_{t+1}^i) + \big(1-\tau_P^i(q_{t+1}^i)\big)q_{t+1}^i, \qquad P_{t+1}^{ji} = (1-\tau_P^j(1-q_{t+1}^i))q_{t+1}^i,$$

and q_{t+1}^i is given by the law of motion

$$q_{t+1}^i = q_t^i + q_t^i(1-q_t^i)\big(\tau_P^i(q_t^i) - \tau_P^j(1-q_t^i)\big).$$

Adopting a state-space notation, the parental policy functions $\tau_P^i(q)$ and $\tau_P^j(1-q)$ must therefore satisfy, for all $q \in (0,1)$,

$$\begin{cases} \tau_P^i(q) = (1-q)\Delta V^i\big(1 + P^{ii}(q') - P^{ji}(q')\big), \\ \tau_P^j(1-q) = q\,\Delta V^j\big(1 + P^{jj}(q') - P^{ij}(q')\big), \end{cases} \tag{14}$$

where $q' = q + q(1-q)\big(\tau_P^i(q) - \tau_P^j(1-q)\big)$, and the transmission probabilities on the right-hand side are evaluated at q' using the same policy functions. This allows us to numerically solve the parental problem through simple iteration of the policy functions.

B.2 Leaders' Equilibrium

We now define the corresponding notion of equilibrium when cultural transmission is mediated by cultural leaders. As in Section 3, leaders maximize the utility of parents in their group. Each leader cares about the next two generations, takes as given the actions of the leader of the other group, and anticipates that a different leader with the same preferences and objectives will choose future socialization levels.

Definition B.2. A leaders' socialization equilibrium at time t, given $q_t^i \in [0,1]$, is $(\tau_t^i, \tau_t^j) \in [0,1]^2$ and $q_{t+1}^i \in [0,1]$ such that, for each $i \in \{a,b\}$, letting $j \neq i$,

$$\begin{aligned} &\tau_t^i \in \arg\max U_t^i + U_{t+1}^i \\ &\text{subject to:}\quad P_t^{ii} = \tau_t^i + (1-\tau_t^i)q_t^i, \qquad P_t^{ij} = (1-\tau_t^i)(1-q_t^i), \\ &\qquad (\tau_{t+1}^i, \tau_{t+1}^j) \text{ satisfy the same equilibrium condition at time } t+1 \text{ given } q_{t+1}^i, \\ &\qquad q_{t+1}^i - q_t^i = q_t^i(1-q_t^i)(\tau_t^i - \tau_t^j), \\ &\qquad \tau_t^j \text{ given.} \end{aligned} \tag{15}$$

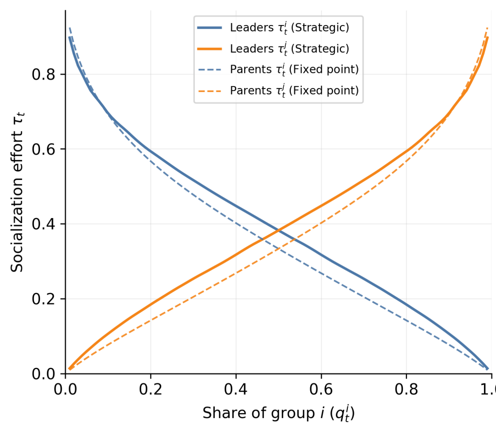
As with the parents let $q_L^i(q_t^i)$ denote the optimal socialization effort of the leader group i when the share of trait i in the population is q_t^i. The first order conditions of the leaders' problem are identical to those in the two-period model, since each leader only cares about the next two generations. The leaders' equilibrium socialization levels are characterized as the fixed point of the system

$$\begin{aligned} \tau_L^i(q_t^i) &= (1-q_t^i)\Delta V^i\Big[1 + \big(P_{t+1}^{ii} - P_{t+1}^{ji}\big)\Big] + P_t^{ii}\frac{\partial P_{t+1}^{ii}\Delta V^i}{\partial \tau_t^i} + (1-P_t^{ii})\frac{\partial P_{t+1}^{ji}\Delta V^i}{\partial \tau_t^i} \\ \tau_L^j(1-q_t^i) &= (1-q_t^j)\Delta V^j\Big[1 + \big(P_{t+1}^{jj} - P_{t+1}^{ij}\big)\Big] + P_t^{jj}\frac{\partial P_{t+1}^{jj}\Delta V^j}{\partial \tau_t^j} + (1-P_t^{jj})\frac{\partial P_{t+1}^{ij}\Delta V^j}{\partial \tau_t^j} \end{aligned} \tag{16}$$

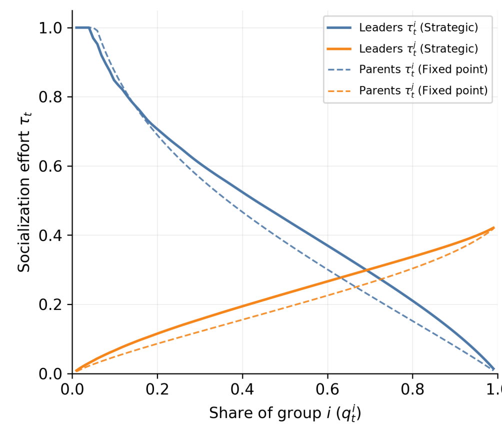
subject to the usual definitions of the transmission probabilities and the law of motion for q^i.[21] All transmission probabilities on the right-hand side are evaluated using the same policy functions $\tau_L^i(\cdot)$ and $\tau_L^j(\cdot)$ at the relevant arguments (q_t^i or q_{t+1}^i). By stationarity, the same system characterizes leaders' socialization choices at any date t.

To numerically estimate the equilibria we iterate on the system of first order conditions by approximating the partial derivatives of the P_{t+1} terms in expression 16 by using finite differences. Figure 7 gives some graphs from the infinite horizon simulations.

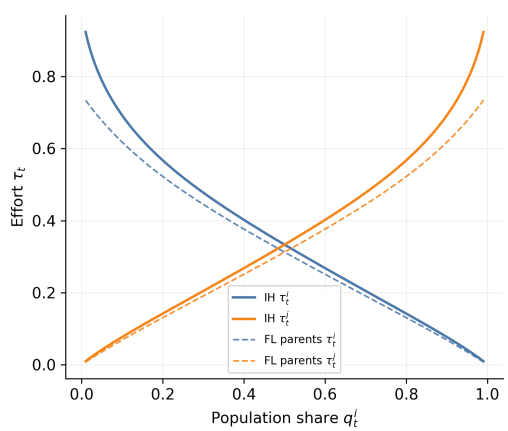
(a) Infinite-horizon leaders versus parents for $(\Delta V^i, \Delta V^j) = (0.50, 0.50)$.



(b) Infinite-horizon leaders versus parents for $(\Delta V^i, \Delta V^j) = (0.60, 0.30)$.



(c) Infinite-horizon parents versus two-period parents for $(\Delta V^i, \Delta V^j) = (0.50, 0.50)$.



(d) Infinite-horizon parents versus two-period parents for $(\Delta V^i, \Delta V^j) = (0.60, 0.30)$.

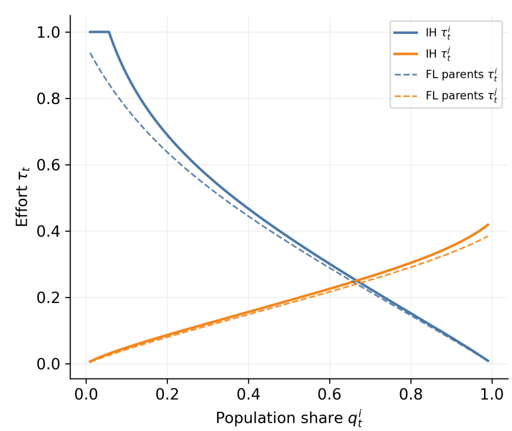


Figure 7: Infinite-horizon simulation comparisons. The top row compares the Markov-perfect socialization choices of cultural leaders and parents; the bottom row compares the parental choices in the infinite-horizon Markov formulation with those in the two-period formulation.

[21] Differentiability of τ_L^i follows from the finite horizon on the preferences and the fact that we restrict the parameters of the model such that an interior solution is always guaranteed.

B.3 Planner's Problem

In the most natural extension to the planner's problem to the one in the two period case, the planner is only able to control the socialization choices for both groups of a given generation, and takes the future generations' choices as given. In the case where the world ended only after two periods, the last period choice was identical across parents and leaders, so a natural choice was to fix that as the choice of the planner as well. In the infinite horizon the issue of the next period choice that the planner anticipates is more intricate.

An interesting notion of efficiency in this context is one given by a sequence of planners. That is, each planner will anticipate that a different planner, with the same objective but for the next-period parents, will pick socialization efforts in the next time period. Formally:

Definition B.3. Efficient socialization at time t, given q_t^i, is $(\tau_t^i, \tau_t^j) \in [0,1]^2$ and $q_{t+1}^i \in [0,1]$ such that for some Pareto weight $0 \leq \alpha \leq 1$:

$$
\begin{aligned}
(\tau_t^i, \tau_t^j) \in \arg \max_{(\tau^i,\tau^j)\in[0,1]^2} \; & \alpha\,(U_t^i + U_{t+1}^i) + (1-\alpha)\,(U_t^j + U_{t+1}^j) \qquad (17)\\
\text{subject to: } & P_t^{ii} = \tau_t^i + (1-\tau_t^i)q_t^i, \qquad P_t^{ij} = (1-\tau_t^i)(1-q_t^i)\\
& P_t^{jj} = \tau_t^j + (1-\tau_t^j)(1-q_t^i), \qquad P_t^{ji} = (1-\tau_t^j)q_t^i\\
& (\tau_{t+1}^i, \tau_{t+1}^j) \in \arg\max(17) \text{ evaluated at state } q_{t+1}^i\\
& q_{t+1}^i - q_t^i = q_t^i(1-q_t^i)\big(\tau_t^i - \tau_t^j\big).
\end{aligned}
$$

The definition restricts the planner in a way such that the weights of the planner for each group remain constant over time.

The following Lemma is a slight generalization of the mechanism behind the parental efficiency result in Proposition 4.1. It states that the effect of changing τ_t^i and τ_t^j on the utility components related to changes in q_{t+1}^i is anti-symmetric.

Lemma B.1. *For any decision rule* $\tau^i(q), \tau^j(q), \frac{\partial P_{t+1}^{ii}}{\partial \tau_t^i} = -\frac{\partial P_{t+1}^{ii}}{\partial \tau_t^j}$ *and* $\frac{\partial P_{t+1}^{ji}}{\partial \tau_t^i} = -\frac{\partial P_{t+1}^{ji}}{\partial \tau_t^j}$.

Proof. From the chain rule we have $\frac{\partial \tau_{t+1}^i}{\partial \tau_t^i} = \frac{\partial \tau_{t+1}^i}{\partial q_{t+1}^i}\frac{\partial q_{t+1}^i}{\partial \tau_t^i}$ (and similarly for τ_t^j). The law of motion implies that $\frac{\partial q_{t+1}^i}{\partial \tau_t^i} = -\frac{\partial q_{t+1}^i}{\partial \tau_t^j}$ and thus $\frac{\partial \tau_{t+1}^i}{\partial \tau_t^i} = -\frac{\partial \tau_{t+1}^i}{\partial \tau_t^j}$. The result follows by expanding P_{t+1}^{ii} and P_{t+1}^{ji} and calculating the derivatives. □

Proposition B.1. *Suppose that in the parental equilibrium* $P_t^{ii}\frac{\partial P_{t+1}^{ii}\Delta V^i}{\partial \tau_t^i} + (1-P_t^{ii})\frac{\partial P_{t+1}^{ji}\Delta V^i}{\partial \tau_t^i}$ *and* $P_t^{jj}\frac{\partial P_{t+1}^{jj}\Delta V^j}{\partial \tau_t^j} + (1-P_t^{jj})\frac{\partial P_{t+1}^{ij}\Delta V^j}{\partial \tau_t^j}$ *have the same sign[this is the analogue to Proposition A.1] and* *the optimal choices of the planner are differentiable. Then parents are generically efficient.*

Proof. Let $\tau_P := (\tau^i_P, \tau^j_P)$ Denote the parental equilibrium choices according to Definition B.1. The planner's first order condition evaluated at that choice with respect to τ^i and τ^j respectively yields:

$$\alpha\left(P^{ii}_t \frac{\partial P^{ii}_{t+1}\Delta V^i}{\partial \tau^i_t} + (1-P^{ii}_t)\frac{\partial P^{ji}_{t+1}\Delta V^i}{\partial \tau^i_t}\right) + (1-\alpha)\frac{\partial}{\partial \tau^i_t}(U^j_t + U^j_{t+1}) \tag{18}$$

$$(1-\alpha)\left(P^{jj}_t \frac{\partial P^{jj}_{t+1}\Delta V^j}{\partial \tau^j_t} + (1-P^{jj}_t)\frac{\partial P^{ij}_{t+1}\Delta V^j}{\partial \tau^j_t}\right) + \alpha\frac{\partial}{\partial \tau^j_t}(U^i_t + U^i_{t+1}) \tag{19}$$

Let $A := P^{ii}_t \frac{\partial P^{ii}_{t+1}\Delta V^i}{\partial \tau^i_t} + (1-P^{ii}_t)\frac{\partial P^{ji}_{t+1}\Delta V^i}{\partial \tau^i_t}, B := P^{jj}_t \frac{\partial P^{jj}_{t+1}\Delta V^j}{\partial \tau^j_t} + (1-P^{jj}_t)\frac{\partial P^{ij}_{t+1}\Delta V^j}{\partial \tau^j_t}$ be the value of these expression evaluated at the equilibrium choices, and note that $\frac{\partial}{\partial \tau^j_t}(U^i_t + U^i_{t+1}) = P^{ii}_t \frac{\partial P^{ii}_{t+1}\Delta V^i}{\partial \tau^j_t} + (1-P^{ii}_t)\frac{\partial P^{ji}_{t+1}\Delta V^i}{\partial \tau^j_t}$ (similarly for j). By Lemma B.1, (18) and (19) become:

$$\alpha A + (1-\alpha)(-B)$$

$$(1-\alpha)B - \alpha A$$

Assuming that A and B have the same sign[22], letting $\alpha = \frac{B}{A+B}$ sets both equations to 0 and therefore shows that parental choices are indeed efficient. □

The result is in fact a bit more general than this: no matter what choice the planner anticipates in the next period, parents are efficient as long as it holds that $AB \geq 0$, and planners only control the choices for one time period.

The proof further implies that the parental choice is only efficient for this specific Pareto weight α. A natural notion of planner to consider in this environment is one where the Pareto weights are given by the share of each culture $q^i_t, 1-q^i_t$ at $t = 0$. Parental socialization choices are trivially not efficient under that definition.

Proposition B.2. *Cultural leaders' equilibrium choices are generically inefficient.*

Proof. The proof is almost identical to the one in Proposition 4.3. Let A, B denote the values of $P^{ii}_t \frac{\partial P^{ii}_{t+1}\Delta V^i}{\partial \tau^j_t} + (1-P^{ii}_t)\frac{\partial P^{ji}_{t+1}\Delta V^i}{\partial \tau^j_t}$ and $P^{jj}_t \frac{\partial P^{jj}_{t+1}\Delta V^j}{\partial \tau^i_t} + (1-P^{jj}_t)\frac{\partial P^{ij}_{t+1}\Delta V^j}{\partial \tau^i_t}$ evaluated at the leaders'

[22]This is exactly the same condition as in Proposition A.1. It states that at the parental equilibrium both groups benefit from an increase in their socialization

equilibrium choices τ_L^*. Then the gradient vectors at τ_L^* are:

$$\nabla W^i(\tau_L^*) = \begin{pmatrix} 0 \\ \\ A \end{pmatrix} \text{ and } \nabla W^j(\tau_L^*) = \begin{pmatrix} B \\ \\ 0 \end{pmatrix}$$

The profitable deviation thus always exists, and the direction of it depends on the signs of A, B. $\square$